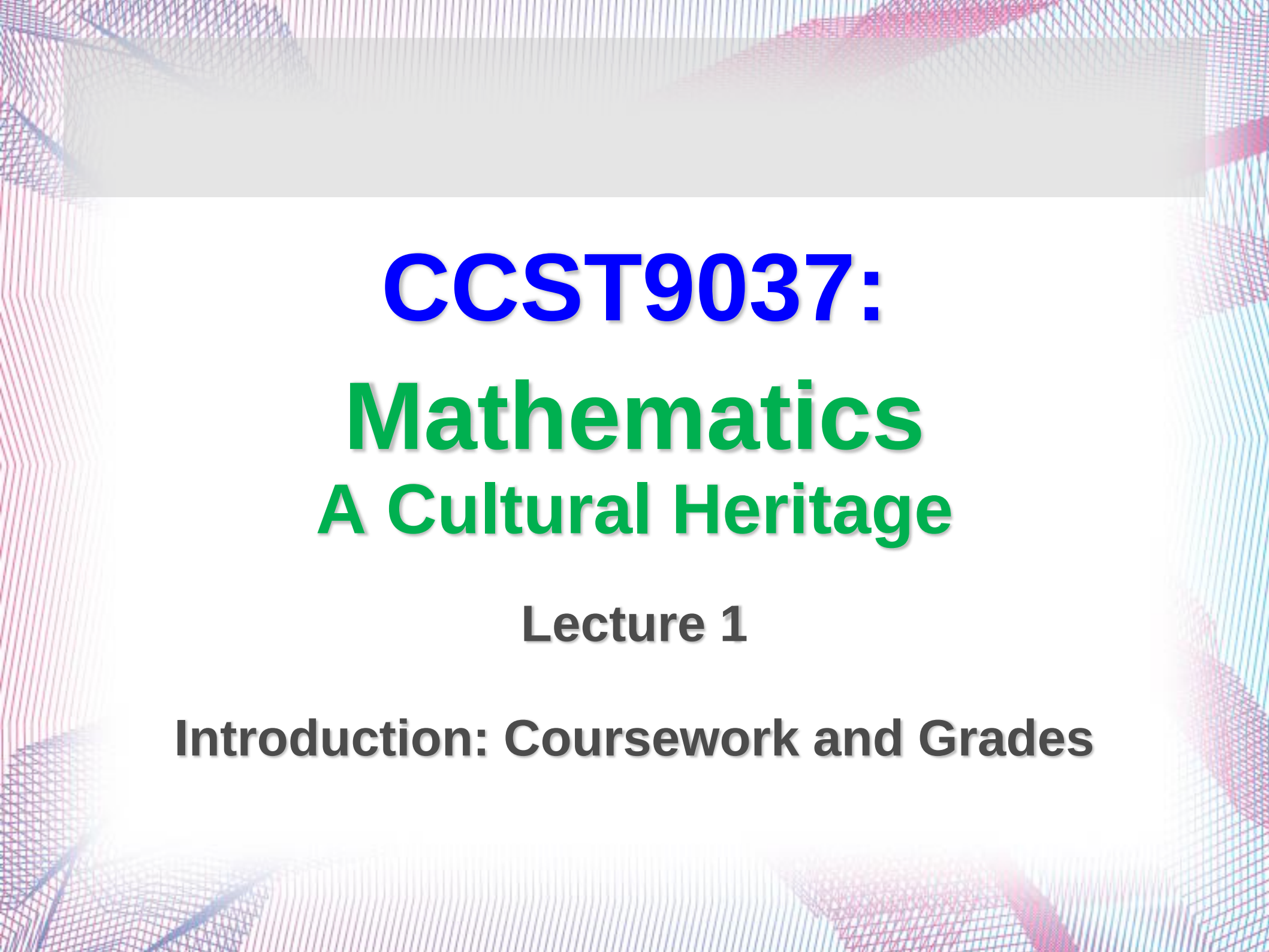




CCST9037:

Mathematics

A Cultural Heritage

Lecture 1

Introduction: Coursework and Grades

Lecture Outline

1. Course learning outcomes
2. Assessment and student work
3. Nature of the course
4. Level of Mathematics required
5. Reading assignment for this week

Course Learning Outcomes

1. Appreciate and describe the beauty, the utility, and the “way” of mathematics.
2. Comprehend and describe how mathematics was and is being developed as a work of human culture.
3. Investigate and describe the interplay between mathematics and other areas of human culture.
4. Investigate and explain the role of mathematics in the development of civilization.

Course Website

1. HKU Portal → My eLearning → Moodle courses
(<https://moodle.hku.hk/course/view.php?id=76757>)



The screenshot displays the Moodle course interface for "CCST9037 Mathematics: A Cultural Heritage [Section 2A, 2016]". On the left is a navigation menu with sections for "Home", "Current course", and "Add Common Functions". The main content area features a header image of a building, followed by buttons to "Add a resource..." and "Add an activity...". Below this is a "Course Information" section listing various course elements like "Default course forum", "News announcement", and "Course Information", each with an "Edit" button and a status icon. At the bottom of the main area are another "Add a resource..." and "Add an activity..." button pair. On the right side, there are several utility boxes: "Exam Timetable", "Search forums" with an input field and "Advanced search" link, "Latest news" with an "Add a new topic..." button and a message that no news has been posted yet, and "Upcoming events" with a message that there are no upcoming events.

Assessment

Task	Weighting
In-class worksheets	10%
Tutorial participation	20%
Written assignments	30%
Essay & presentation	40%

Other obligations

1. Reading assignments
2. Watching videos
3. Viewing/interacting with website content

Analyzing all of the above in the context of this course.

Lectures

1. **Hybrid** synchronous/asynchronous
2. Recorded videos (asynchronous)
3. Weekly Zoom meetings/discussion/question and answer sessions (synchronous)
4. All synchronous materials will be recorded.

Tutorials

1. Attendance **Compulsory**
2. Small class size
3. Objective and focus:
 - Discussion / Interaction
 - Working on problems
 - Presenting your view/answers/solutions
 - Raising questions

Tutorial logistics

1. Tutorials start on Wednesday, **Sept. 15, 2021**.
2. **No** tutorials during the reading week.
3. Please note that we are using the University's Tutorial Sign-up Application for enrolment into your tutorial group; see:

<http://moodle-support.hku.hk/tutorial-sign-application-student>

In essence you can pick your tutorial via My E-learning tab after you log into HKU Portal, or you can use the corresponding mobile app developed by the University.

Assignments (Brief Overview)

1. **Two solo** written assignments
2. Work on your own
3. Typical length **500-850** words
4. Expect **at least** 1.5 weeks to complete
5. Comparison and contrast to reading assignments part of the basis of assignment
6. **Group project**: in depth (**2000-3000** words)
7. **Group Presentations**: 2-3 students

Grade determination

Grade	Requirement
A	Demonstrate thorough knowledge and skills required for attaining all the course learning outcomes. Show strong analytical and critical abilities and logical thinking. Apply highly effective organizational and presentational skills.
B	Demonstrate substantial command of knowledge and skills required for attaining at least most of the course learning outcomes. Show evidence of analytical and critical abilities and logical thinking. Apply effective organizational and presentational skills.
C	Demonstrate general but incomplete command of knowledge and skills required for attaining most of the course learning outcomes. Show evidence of some analytical and critical abilities and logical thinking. Apply moderately effective organizational and presentational skills.
D	Demonstrate partial but limited command of knowledge and skills required for attaining some of the course learning outcomes. Show evidence of some coherent and logical thinking, but with limited analytical and critical abilities. Apply limited or barely effective organizational and presentational skills.
F	Demonstrate little or no evidence of command of knowledge and skills required for attaining the course learning outcomes. Lack of analytical and critical abilities, logical and coherent thinking. Organization and presentation skills are minimally effective or ineffective.

Acknowledgements

1. This Common Core course is based on a previous course, with the same course title, developed and taught by Professor Man-keung Siu (蕭文強教授).
2. Dr. Nam-Kiu Tsing was one of the originators of this incarnation of the course.

Caveats

1. This is not a “regular” mathematics course (i.e., Calculus, Linear Algebra, etc.)
2. The goal and focus is not solving numerical problems or proving theorems
3. The main work involves **reading, writing, analyzing, discussing, and debating** about how mathematics plays a role in human society and culture and **presenting** own's views.

Role of Mathematics in Human Civilization

1. Foundation of science/technology
2. Strong presence in wide range of other areas
3. One of the major threads (together with language, science, and the arts) that form the core of human civilization

Role of Mathematics in Human Civilization

“... Mathematics sets the standard of objective truth for all intellectual endeavours; science and technology bear witness to its practical usefulness. Besides language and music, [mathematics] is one of the primary manifestations of the free creative power of the human mind, and it is the universal organ for world-understanding through theoretical construction. Mathematics must therefore remain an essential element of the knowledge and abilities which we have to teach, of the culture we have to transmit, to the next generation.”

- Hermann Weyl



"Hermann Weyl ETH-Bib Portr 00890" by ETH Zürich - ETH-Bibliothek Zürich, Bildarchiv. Source: Wikipedia

Course Outline

1. Study examples gathered from the long history of humankind, around our daily lives, in diverse areas of human activities, so that we will be able to comprehend how mathematics was, and is being, developed as a work of human endeavor with cultural, intellectual, and social contexts.
2. Investigate the role of mathematics in the development of other areas of our civilization
3. Examine the interplay between mathematics and other cultural pursuits such as
 - philosophy
 - the arts
 - science and technology
 - ...and how they have affected each others' development

Level of Mathematics Required

1. Very minimal!
2. You should have intellectual curiosity and be willing to participate in the logical process of reasoning.
3. You can drive a car without understanding how it works (unless the car breaks – we need mechanics, too!).

Reading Assignment

- Read article:
“Numbers: Easy as 1, 2, 3 – people come into the world ready to count its wonders”, in The Economist, Dec 30, 2008
- URL: www.economist.com/node/12847128
- (may require registration with The Economist, which is free)